

C. Spy Syndrome 2

time limit per test: 2 seconds
memory limit per test: 256 megabytes

After observing the results of Spy Syndrome, Yash realised the errors of his ways. He now believes that a super spy such as Siddhant can't use a cipher as basic and ancient as Caesar cipher. After many weeks of observation of Siddhant's sentences, Yash determined a new cipher technique.

For a given sentence, the cipher is processed as:

1. Convert all letters of the sentence to lowercase.
2. Reverse each of the words of the sentence individually.
3. Remove all the spaces in the sentence.

For example, when this cipher is applied to the sentence

Kira is childish and he hates losing

the resulting string is

ariksihsidlihcdnaehsetahgnisol

Now Yash is given some ciphered string and a list of words. Help him to find out any original sentence composed using only words from the list. Note, that any of the given words could be used in the sentence multiple times.

Input

The first line of the input contains a single integer n ($1 \leq n \leq 10\,000$) — the length of the ciphered text. The second line consists of n lowercase English letters — the ciphered text t .

The third line contains a single integer m ($1 \leq m \leq 100\,000$) — the number of words which will be considered while deciphering the text. Each of the next m lines contains a non-empty word w_i ($|w_i| \leq 1\,000$) consisting of uppercase and lowercase English letters only. It's guaranteed that the total length of all words doesn't exceed $1\,000\,000$.

Output

Print one line — the original sentence. It is guaranteed that at least one solution exists. If there are multiple solutions, you may output any of those.

Examples

input	Copy
30 ariksihsidlihcdnaehsetahgnisol 10 Kira hates is he losing death childish L and Note	
output	Copy
Kira is childish and he hates losing	

input	Copy
12 iherehtolleh 5 HI Ho there HeLlo hello	
output	Copy
HI there HeLlo	

Note

In sample case 2 there may be multiple accepted outputs, "HI there HeLlo" and "HI there hello" you may output any of them.

Manthan, Codefest 16

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
362136673	Feb/09/2026 17:00	Accepted
362136373	Feb/09/2026 16:58	Accepted
362135597	Feb/09/2026 16:53	Compilation error
362134836	Feb/09/2026 16:48	Time limit exceeded on test 67

→ Problem tags

data structures dp hashing implementation sortings string suffix structures strings *1900

No tag edit access

→ Contest materials

Announcement (en)

Tutorial (en)